Arista Access Node ELEC-ACC-01 Optical Transport Failure SOP

Node: ELEC-ACC-01 | **Vendor:** Arista | **Severity:** Major | **Alarm Code:** ALARM_5319

Description

This Standard Operating Procedure (SOP) addresses the failure domain of Optical Transport Failure on the Arista Access Node ELEC-ACC-01 at the Electronic City Hub. The failure domain encompasses SFP/XFP degradation, fiber cuts, and Rx/Tx power loss. The underlying protocol or hardware mechanics involve the Arista EOS (Extensible Operating System) handling optical transport failures through the use of SFP/XFP modules, which convert optical signals to electrical signals for processing by the Arista switch. When an SFP/XFP module degrades or a fiber cut occurs, the optical signal is either attenuated or completely lost, resulting in a failure to transmit or receive data. This failure domain is critical as it impacts the overall network availability and reliability.

Pre-requisite Checks

- pre-requisite: Verify the site location of Electronic City Hub and confirm the node ID as ELEC-ACC-01.

- pre-requisite: Check the redundant link status to ensure it is operational and can take over in case of a failure.

- pre-requisite: Dispatch remote hands to Electronic City Hub to assist with the troubleshooting and resolution process.

Potential Root Causes

Cause	Probability
SFP/XFP degradation	High
Fiber cut	Medium
Rx/Tx power loss	Low

Diagnostic Commands & Expected Outputs

```
Command: show interfaces fiber e1/1, Expected Output: Interface e1/1, Hardware: arista-fiber 10GBF, Description: Fiber Link to ELEC-ACC-02, Description: This command isolates the fiber interface of the Arista Access Node ELEC-ACC-01 and displays its description, which includes the type of fiber link and its connection to another node (ELEC-ACC-02).

Command: show interfaces fiber e1/1 statistics, Expected Output: Interface e1/1, Statistics: 1234567890 bytes received, 1234567890 bytes sent, 0 errors, 0 drops, Description: This command displays the statistics of the fiber interface, including the number of bytes received and sent, as well as any errors or drops. This can help identify potential issues such as fiber degradation or cuts.
```

Resolution Steps

Step 1: Step: Replace SFP/XFP module, Command: install slot 1 10GBF-SFP-00, Description: Replace the SFP/XFP module in slot 1 with a new one, which will restore the optical transport functionality.

Step 2: Step: Verify fiber integrity, Command: show interfaces fiber e1/1, Description: Verify that the fiber interface is operational and the link is up, which ensures that the fiber cut or degradation has been resolved.

Post-Check Verification

```
Command: show interfaces fiber e1/1, Expected Output: Interface e1/1, Hardware: arista-fiber 10GBF, Description: Fiber Link to ELEC-ACC-02, Description: Verify that the fiber interface is operational and the link is up, which ensures that the fiber cut or degradation has been resolved.

Command: show interfaces fiber e1/1 statistics, Expected Output: Interface e1/1, Statistics: 1234567890 bytes received, 1234567890 bytes sent, 0 errors, 0 drops, Description: Verify that the fiber interface is operational and there are no errors or drops, which ensures that the SFP/XFP module replacement or fiber integrity verification has been successful.
```

Roll-back Procedures

- Step: Revert SFP/XFP module replacement, Command: remove slot 1 10GBF-SFP-00, Description: Revert the SFP/XFP module replacement by removing the new module and reinserting the original one, which will restore the failed configuration.

- Step: Verify fiber integrity, Command: show interfaces fiber e1/1, Description: Verify that the fiber interface is operational and the link is up, which ensures that the fiber cut or degradation has been resolved.